

AJAY KUMAR RAY

Madhubani, Bihar

☎ [+91-9693984515](tel:+91-9693984515) ✉ rayajayk@gmail.com [in](#) [Linkedin](#) [G](#) [Github](#)

EDUCATION

Central University of Jammu

B.Tech - Electronics and Communication Engineering

CGPA : 9.00 (current)

2022 - 2026

Jammu, J&K

Jagdish Nandan College (BSEB)

12th Standard Percentage : 75.8%

2019 - 2021

Madhubani, Bihar

MKU GOVT Vidyalaya (BSEB)

10th Standard First Division

2018 - 2019

Madhubani, Bihar

RELEVANT COURSEWORK

- Electronic Devices & Circuits
- CMOS Digital VLSI Design
- Control Systems
- Cyber-Physical Systems
- Digital Electronics
- IoT

EXPERIENCE

Intern – E-Tarangini Innovations Pvt. Ltd.

January - March, 2026

Role - System Developer

Jammu

- Developed a LoRa-based wireless communication system for real-time sensor data transmission.
- Simulated industrial monitoring concepts similar to SCADA systems.
- Integrated DHT11, Ultrasonic, and LDR sensors using Raspberry Pi and GPIO interfacing.
- Implemented UART communication and structured data transfer using JSON format.
- Enabled cloud-based monitoring by integrating the system with ThingSpeak platform.

Internship Trainee – Bharat Electronics Limited (BEL)

Aug 2025 - Dec 2025

Role - Intern

Panchkula, Haryana

- Worked on the LINK-II Tactical Data Link System, enabling secure, real-time data exchange between airborne, naval, and ground defence platforms.
- Analyzed end-to-end system architecture and data flow across multiple LRUs, including RF modem, encryption, and Ethernet-based interconnections.
- Studied TDMA-based communication and time synchronization using 1-PPS and 10 MHz reference clocks for reliable, collision-free transmission.
- Observed system integration, testing, and validation procedures followed for mission-critical defence communication systems.

Workshop on Robot Operating System (ROS)

April 2024

Role - Workshop Attendee

CUJammu

- Gained hands-on experience with the ROS framework, focusing on its architecture, communication mechanisms (nodes, topics, services), and package management.
- Developed and simulated basic robotics applications using ROS, including robot motion planning and sensor integration.
- Implemented ROS-based control systems to manage sensor data and automate robot behavior in simulated environments.

PROJECTS

All India Model Rocketry Competition (ISRO)

- Selected and qualified for the prestigious national-level rocketry competition organized by ISRO and IN-SPACe.
- Designed and implemented a 1 kg payload ejection system at 1000 m altitude.
- Developed and optimized parachute-based recovery mechanisms for safe descent.
- Ensured structural integrity and aerodynamic stability during payload ejection.
- Conducted simulations and real-world testing to validate deployment and recovery efficiency.

Object Detection Radar System

- Built and tested an Arduino-based radar system to detect objects and visualize their positions in real-time.
- Integrated an ultrasonic sensor and servo motor to scan the environment across a range of angles (15°–180°).
- Developed a radar-like interface using Processing4 to display object positions dynamically on screen.
- Implemented real-time data acquisition, processing, and visualization for effective object detection and tracking.

Common Emitter BJT Amplifier – Design and Testing

- Designed and implemented a common-emitter BJT amplifier on hardware
- Performed biasing calculations to ensure stable amplifier operation
- Measured voltage gain and frequency response using CRO/function generator

TECHNICAL SKILLS

Languages: Verilog, VHDL, Python, C++

Hardware & Embedded : Arduino, Raspberry Pi, Sensors, Embedded Systems

Design Concepts: RTL Design, FSM, ALU, Flip-Flops, Counters, Memory Modules

Tools and Frameworks: VS Code, Git, GitHub, TinkerCAD Circuits, Canva

CERTIFICATIONS

- CMOS Digital VLSI Design - NPTEL
- Semiconductor Devices and Circuits - NPTEL
- Cryogenic Electronics for Quantum Computing - NPTEL
- VLSI SoC Design using Verilog HDL - Maven Silicon
- Internet of Things - Technophilia

EXTRACURRICULAR

- **President, Electronics and Robotics Club** - Organized technical competitions and mentoring sessions.
- **IEEE Student Member** - Engaged in research discussions and professional networking.
- **Commentator, Inter -College Cricket Match** - Provided live match commentary.
- **Organized an event for juniors** - Conducted engaging technical activities and competitions for first and second-year students.